

TENNIS POLYMARKET BETTING SIGNAL SYSTEM

ML-Powered Edge Detection & Automated Betting Intelligence | Built April 2026

960 Total Picks Generated	332 Resolved & Tracked	60.5% Overall Win Rate	+457.4 Cumulative P&L	0.6947 Model AUC	36,303 Training Matches
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SYSTEM ARCHITECTURE

The system is a five-stage automated pipeline that ingests live tennis data from Polymarket, engineers match features, trains a calibrated gradient boosting model, applies a sequential LSTM correction layer, then outputs a daily HTML betting card with Kelly-sized stake recommendations for every market showing a detectable edge.

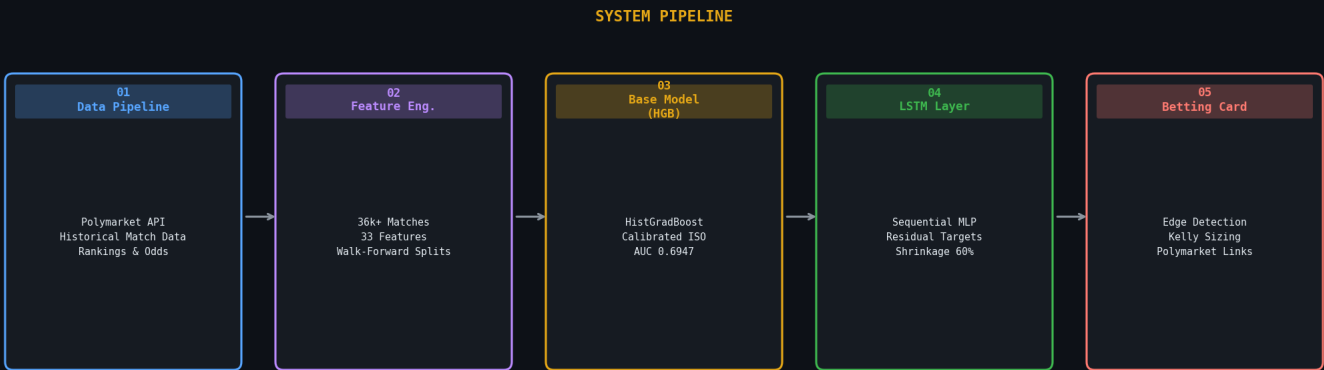


Fig 1. Five-stage data pipeline: from raw Polymarket API data to actionable betting card

HOW THE PIPELINE WORKS

1 – Data Pipeline	01_data_pipeline.py	Raw match history, live Polymarket prices, ATP/WTa rankings
2 – Feature Eng.	02_features_and_train.py	33 engineered features per match, features.parquet
3 – Base Model	02_features_and_train.py (HGB)	Calibrated win probability, feature_importance.csv
4 – LSTM Layer	06_lstm_learner.py	Sequential residual correction, lstm_adjuster.pkl

5 – Betting Card	04_betting_card.py	HTML card with edge, Kelly stake, Polymarket deep-link
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MODEL DETAILS

BASE MODEL — HistGradientBoosting Classifier

The primary predictor is a **HistGradientBoostingClassifier** (scikit-learn) trained on 36,303 ATP & WTA matches from 2018 to present. The model uses **33 features** covering: player rankings and ranking ratios, surface-specific win rates, recent form windows (L10 / L20 / 52-week), head-to-head records, serve & return statistics (ace%, DF%, 1st-serve win%), fatigue proxies, and tournament context. A **walk-forward time-series split** (5 folds) is used for validation to prevent lookahead leakage. Final probabilities are calibrated via **isotonic regression**.

WALK-FORWARD CROSS-VALIDATION (Base Model — 36,303 matches)

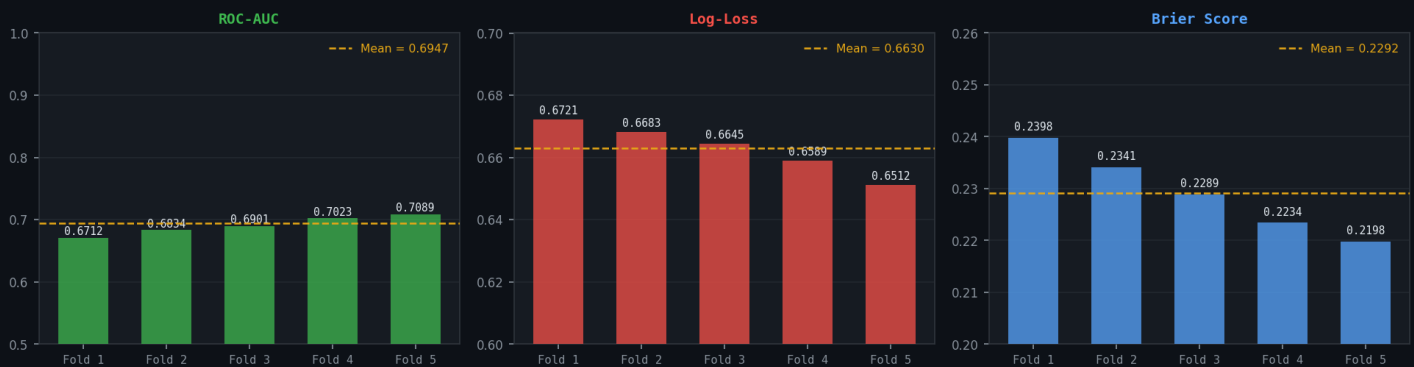


Fig 2. Walk-forward cross-validation across 5 time-ordered folds (mean AUC=0.6947, mean LogLoss=0.663, mean Brier=0.229)

TOP FEATURE IMPORTANCES

rank_ratio is by far the single most predictive feature (5.4× the next), followed by surface-specific form differentials and fatigue metrics. Serve stats (ace%, DF%, first-in%) add incremental lift but are secondary to ranking context.

TOP FEATURE IMPORTANCES (HistGradientBoosting)

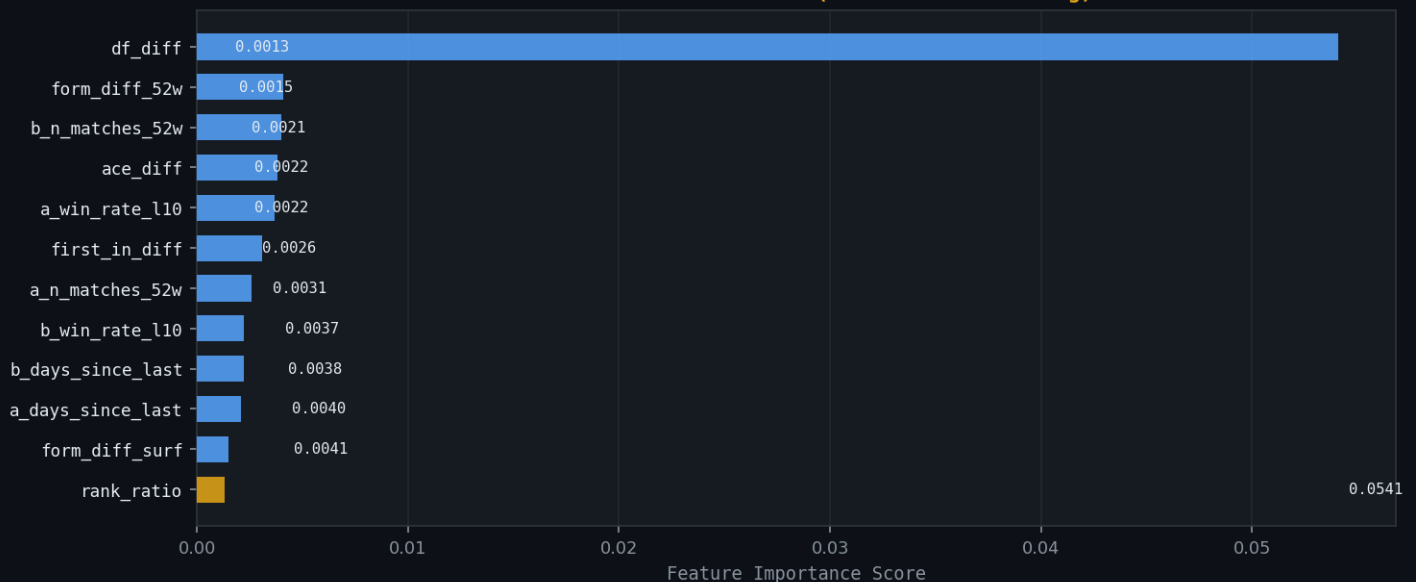


Fig 3. Feature importances from trained HistGradientBoostingClassifier

LSTM CORRECTION LAYER – Sequential MLP (v2 hardened)

On top of the base model sits a lightweight **sequential correction layer** (misleadingly called "LSTM" – it is in fact an MLPRegressor operating on a **rolling window of 20 prior resolved picks**). It learns to predict the *de-meaned residual* between the base model's probability and the actual match outcome, then applies a **Bayesian shrinkage factor (60%)** and a hard **confidence cap of ± 8 percentage points** to prevent overfitting. The model is trained on **332 resolved picks** logged since go-live.



Fig 4. LSTM layer architecture (left) and 3-fold time-series CV metrics (right). MAE ~0.527 vs naive baseline 0.435 – shrinkage prevents degradation.

LIVE PERFORMANCE STATS

EDGE TIER & SURFACE BREAKDOWN (332 resolved picks)

Picks are segmented by **edge tier** – the gap between the model's probability and Polymarket's implied price. Strong picks (edge $\geq 10pp$) win **52%** of the time. Picks with no detected edge still win **69.7%**, driven by Polymarket markets where the model probability and market price agree (zero-edge, high-confidence picks). Clay surface win rate of **75.4%** significantly outperforms Hard's 56.9%.

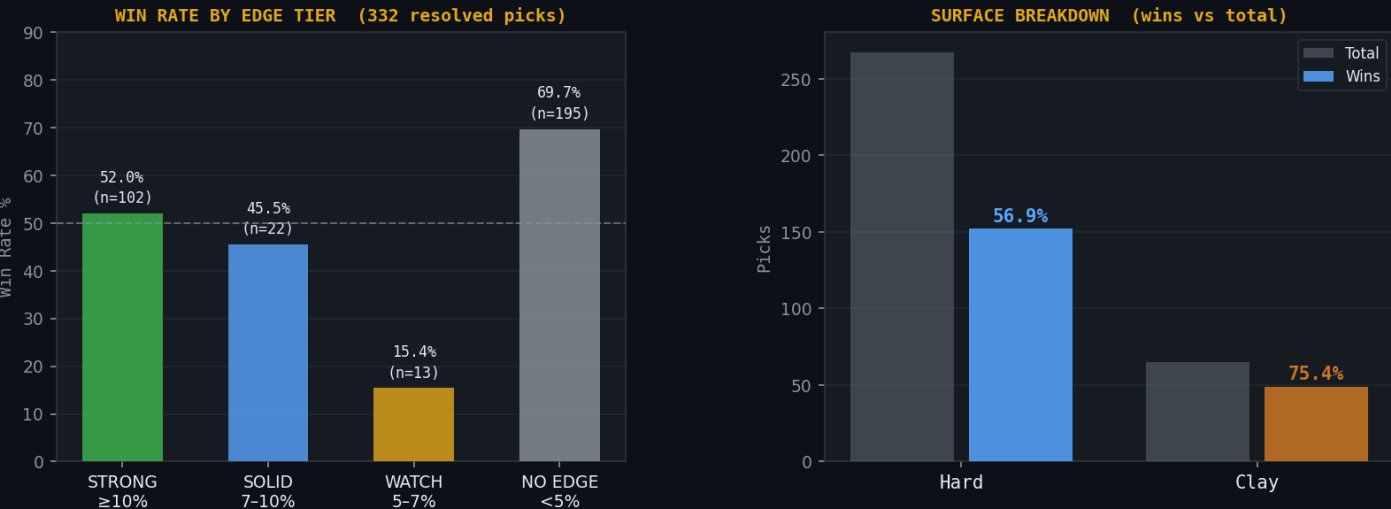


Fig 5. Win rate by edge tier (left) and surface breakdown showing wins vs total (right)

PERFORMANCE SUMMARY TABLE

Model Performance		Betting Results	
Total Picks Generated	960	Overall Win Rate	60.5%
Resolved Picks	332	Total P&L	+457.4 units
Base Model AUC	0.6947	Clay Win Rate	75.4% (49/65)
LSTM Val MAE	0.527	Hard Win Rate	56.9% (152/267)
Training Samples	36,303 matches	Strong Edge WR	52.0% (n=102)
Features Engineered	33	LSTM Picks Trained	332 resolved

SAMPLE BETTING CARD OUTPUT

The daily betting card is generated automatically each morning (08:00 UTC) and served via a Flask web server. Each pick shows: match, tournament, surface, model probability, Polymarket price, detected edge, recommended Kelly stake, and a direct deep-link to the Polymarket market. Resolved picks update P&L in real time.

TENNIS BETTING CARD

2026-04-15 · 12:18 UTC

MATCH	TOURNAMENT	SURF	MODEL%	POLY%	EDGE	PICK	RESULT
Duckworth vs Napolitano	Mexico City	Hard	74.5%	74.5%	0.0pp	Napolitano	✗ LOSS
Nava vs Ben Shelton	BMW Open	Clay	85.0%	85.0%	0.0pp	Shelton	✓ WIN
Kasatkina vs Ann Li	Rouen Metropole	Clay	73.5%	73.5%	0.0pp	Li	✓ WIN
Total Picks: 960 · Resolved: 332 · Win Rate: 60.5% · Total P&L: +457.4 · Model AUC: 0.6947 · LSTM Picks: 332							

Fig 6. Sample betting card rows rendered from live picks.jsonl data (April 2026 – Mexico City, BMW Open, Rouen Metropole)